

Is a Biology Degree Worth It in the AI Era?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

5.0

AI EXPOSURE · CLINICAL TRIALS / FIELD RESEARCH
where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in a life sciences degree, or out of one and into the job hunt. So this guide skips the "should you study biology" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where life sciences sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. Already in it is not too late — not close. Life sciences isn't one destination, it's a wide spread of them, from 4.2 to 7.8, and almost none of your exposure is fixed yet. Where you land inside that range is decided mostly by moves still in front of you — chiefly, whether you keep your hands and steer toward regulated work. The second half of this guide is those moves.

And one thing specific to life sciences. The AI finding here is genuinely surprising and the financial finding is genuinely important, and they're different things.

The surprise: the computational, tech-forward end of biology is *more* exposed than the hands-on laboratory end. The important part: a biology bachelor's on its own leads to technician work at around \$47,000, while the same field in industry averages three times that. The section on the financial picture below is the one to read carefully — where you aim inside that range determines whether the numbers work.

The short answer

Yes for hands-on and regulated work. Much less so for the computational end — which is the opposite of what almost everyone assumes.

Field ecology and environmental science scores **4.2** out of 10. Wet lab and bench research scores **5.2**. Bioinformatics and computational biology scores **7.1**.

The people at the bench are better protected than the people analyzing what the bench produces.

And the bigger question for most students isn't exposure at all — it's that a life sciences bachelor's is a gateway qualification rather than a destination, and where it leads varies by a factor of three in pay. Where you steer from here is decided mostly by moves still in front of you, which is the whole reason being already in it is not a problem.

The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Field ecology & environmental science	3.6	3.9	4.2	+0.6	Low–Moderate
Clinical trials & regulatory affairs	3.9	4.3	4.6	+0.7	Moderate
Wet lab & bench research	4.4	4.8	5.2	+0.8	Moderate
Lab technician & diagnostics	4.9	5.4	5.8	+0.9	Moderate–High
Bioinformatics & computational biology	5.6	6.4	7.1	+1.5	High
Scientific data curation & literature analysis	6.4	7.2	7.8	+1.4	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, entry data analysis scores 8.2, licensed engineering 4.0, and bedside nursing 2.8.

Read the table this way, and note the ordering. The most technically sophisticated, best-paid and most competitive track in this list is also the most exposed. That inversion is the whole guide. Your job, from here, is to move toward the protected, regulated end — not to hope the table moves.

The inversion: why computational biology is more exposed than the bench

This is the finding worth carrying beyond biology, because the same pattern appears in agriculture, construction and several other fields in this index.

The intuition most people have is that the tech-forward version of a career must be the future-proof one. Bioinformatics is computational, quantitative, well-paid and hard to get into. Wet lab work is manual, repetitive and lower-status. Surely the first is safer.

The framework says the opposite, and the reason is structural.

Ask what each job actually consists of.

Bioinformatics takes structured data — sequences, expression matrices, imaging output — and produces analysis and code. Structured inputs, computational outputs, no physical presence, no licensed accountability. That is the exposed profile in every field we score, and it doesn't stop being the exposed profile because the subject matter is biology.

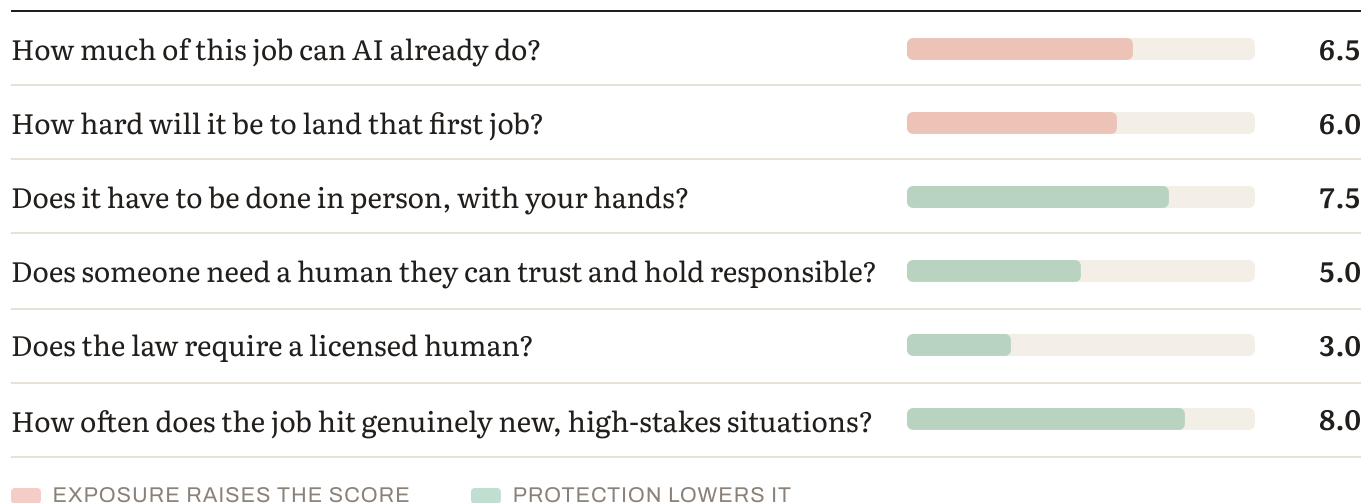
Wet lab work requires hands. Pipetting, cell culture, animal handling, sample preparation, running an instrument that has developed a fault nobody documented. It happens in a physical space, on physical material, and the thing that goes wrong is usually not in any protocol.

So computational biology is exposed for the same reason data analysis is exposed — because it is data analysis. The biology is the subject; the shape of the work is computation.

The general lesson: proximity to technology is not protection. We make this point in the computer science and data science guides too, and life sciences is the clearest demonstration because the two tracks sit inside one degree. Same students, same department, opposite exposure.

One important qualification. This is a statement about the *analysis* layer, not about scientists. A computational biologist who designs the experiment, decides what question is worth asking and is accountable for the interpretation is doing protected work. One who runs pipelines on other people's data is not. The distinction is judgment and ownership, exactly as everywhere else in this index — and it's the distinction you can still move yourself across.

Why the bench holds — the six factors



Here's how wet lab and bench research answers them. Knowing *why* the bench holds at 5.2 while the computational end sits at 7.1 is what tells you which way to move.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The analysis, the literature work and the writing. Not the experiment.

Sequence and structural analysis. Statistical analysis and visualization. Literature review and synthesis. Protocol drafting. Manuscript preparation. Image quantification.

And genuinely: hypothesis generation from existing data is now meaningfully assisted, which is a real change to how research is done.

What resists: doing the experiment. Noticing that a culture looks wrong before any measurement says so. Troubleshooting an instrument. Handling animals. And the tacit craft that makes one person's results reproducible and another's not. Those are the things to be building toward.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Moderate, and this is where the honest problems sit — not with exposure but with pay and progression. The financial picture below covers it.

The way through isn't to take the first technician seat the degree points at — that tier is modestly paid and slow to progress without further qualification. It's to aim *past* it: at regulated and industry work (clinical trials & regulatory affairs 4.6, quality, GMP manufacturing, diagnostics) that pays far better and is protected by named accountability, with real hands-on experience behind you and a working theory of the next step. That's a set of concrete moves, not a hope, and the career-hunting checklist later in this guide lays them out.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Substantially, for bench and field work. Not at all for computational work. That single difference does most of the work in the score table.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

In regulated environments, significantly. Clinical trials, GMP manufacturing and diagnostics all require named responsible individuals, and that accountability is a real protection. In academic research it is weaker. Which means the fastest way to become protected is to get closer to work someone signs off.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

In regulated work, yes — and this is under-appreciated. Clinical trial conduct, GMP manufacturing and clinical diagnostics operate under regulatory frameworks requiring qualified persons. In basic research, essentially none.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly in research — novelty is the entire point of the work — though the stakes of any individual experiment are usually low.

The financial picture, and it matters more than the score

This is the section that should shape what you do next.

A life sciences bachelor's on its own leads to technician work.

- Biological technicians earn a median of around **\$47,000–\$48,000**
- Entry-level lab technician and research assistant roles typically run **\$40,000–\$55,000**
- Clinical research coordinators earn a median of about **\$52,880**
- Environmental scientists do better, at a median near **\$80,000**
- Forensic science technicians earn around **\$63,740**, with 11% projected growth

Now compare the same field in industry.

National average base salaries for life sciences professionals in industry sit in the **mid \$140,000s to low \$160,000s**. Process engineers and manufacturing science specialists earn **\$110,000–**

\$160,000. Senior quality and compliance professionals earn **\$90,000–\$140,000**, with heads of quality exceeding \$180,000. Commercial roles start at **\$110,000–\$170,000** base.

That is a gap of roughly three times between where a biology degree most commonly lands and where the same field pays best.

The gap is not about intelligence or ability. It's about sector. Academic and bench-adjacent roles are funded by grants and university budgets. Industry roles are funded by products. The science overlaps considerably; the economics do not.

And the growth numbers are respectable rather than exciting: medical scientists around 9%, environmental scientists 8%, biological technicians 7%, biochemists and biophysicists 6%, postsecondary biology teachers 5%. Life sciences overall around 5% through 2032, with biotech adding over 20,000 positions in 2025.

Two practical conclusions follow.

The degree is a gateway, not a destination. Most people who do well from a life sciences bachelor's do something else afterwards — medicine, dentistry, veterinary, physician associate, a specialist master's, or a move into industry, regulatory affairs or commercial roles.

And the PhD question needs answering honestly. Doctoral qualifications show a 40–50% salary uplift over bachelor's in BLS classifications, at the cost of four to seven years. That is a genuine return in industry and a much weaker one in academia, where the number of permanent research positions is far smaller than the number of people trained for them.

What the trend shows

Wet lab and bench research: 4.4 → 4.8 → 5.2. Up 0.8 in three years.

Bioinformatics: 5.6 → 6.4 → 7.1. Up 1.5 — nearly double the pace.

The gap has inverted further. In 2023 the difference was 1.2 points; it's now 1.9. The computational end is not converging with the bench end. It's separating from it, in the direction nobody expected. Which is why drifting toward the computational end because it looks more advanced is the one move to avoid.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Real for bench and field work	Durable for now	Lab automation is real but expensive and inflexible; unstructured troubleshooting resists.
Regulatory / licensing	Strong in regulated environments	Most durable	Clinical trials, GMP and diagnostics require named responsible persons.
Judgment under novelty	Strong in research	Eroding at the analysis layer	Hypothesis generation is now assisted; experimental design and interpretation less so.
Human trust / accountability	Real in regulated roles	Durable	Someone signs off on a batch, a trial, a diagnostic result.

The honest read. The protected version of a life sciences career is the regulated one: clinical trials, quality, manufacturing, diagnostics. That is also where the pay is. The two findings point the same way, which is unusual and worth noticing — and it's why the direction you steer matters more than where you happen to start.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Running the analysis pipeline	Directing it, then interrogating the output
Literature review	Synthesized automatically; the scientist judges relevance
Drafting the manuscript	Edited rather than written
Generating hypotheses from data	Increasingly assisted — a genuine change
Doing the experiment	Unchanged
Noticing the culture looks wrong	Unchanged
Signing off a regulated batch or trial	Unchanged

One development worth naming honestly. Hypothesis generation from existing data is the first genuinely intellectual part of science to be meaningfully assisted, and the profession has not settled what that means. Designing the right experiment and interpreting an unexpected result remain human. Proposing candidate explanations increasingly is not. Which means the scientists who go out of their way to build experimental and interpretive judgment — not just produce the analysis — have an edge that's about to get scarcer.

Human strengths that will matter most

1. **Doing the experiment well** — the tacit craft that makes results reproducible. The least automatable and least teachable thing here. 2. **Noticing something is wrong** before any measurement says so. 3. **Regulated accountability** — being the named person who signs off a batch, a trial or a diagnostic result. 4. **Experimental design** — deciding what would actually answer the question. 5. **Interpreting the unexpected** — the result that doesn't fit, which is where most real discovery lives.

Notably absent: analysis pipeline fluency, literature recall and figure production. These were traditional markers of a strong graduate researcher and are the fastest-depreciating. If your current value is mostly on that second list, the work is to shift it onto the first.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The specific tools — the analysis pipeline, the sequencing platform, the imaging software — will turn over many times across a career. The scientists who compound are the ones who stay genuinely curious about how living systems work and re-tool fastest, rather than defending the one method they trained on. In a field where the analysis layer is automating fastest, that flexibility is itself a form of protection — and increasingly it's what employers are hiring for: the trajectory toward experimental judgment and regulated accountability, not the current pipeline fluency.

Other things to consider about this job

What's genuinely good about it

The work is genuinely interesting, and people mean it. Research is one of very few careers where the point is to find out something nobody knew.

- **Growth is steady across most specialisms** — 9% for medical scientists, 8% for environmental scientists, 11% for forensic technicians

- **Industry pay is strong** — mid \$140,000s to low \$160,000s average base, with manufacturing, quality and commercial roles all well above the academic equivalent
- **Biotech added over 20,000 positions in 2025**
- **The degree opens an unusually wide set of doors** — medicine, dentistry, veterinary, physician associate, public health, and industry
- **Environmental and field roles** combine science with outdoor work and score well at 4.2

And regulated work is the under-considered good answer. Regulatory affairs, quality assurance and clinical trial management pay well, are protected by named accountability, and almost nobody applies to them from a biology degree.

What's genuinely hard about it

The bachelor's alone leads to modestly-paid technician work, and this is the dominant practical problem. Median around \$47,000, with limited progression without further qualification.

The academic pipeline is the honest difficulty. Far more people are trained for research careers than there are permanent research positions, and the postdoctoral stage can extend for years on short contracts. Anyone considering an academic path should understand the numbers before committing four to seven years.

Funding is grant-dependent in academia, which makes job security genuinely uncertain in a way industry roles are not.

And the entry-level market is competitive. Life sciences graduates are numerous, and the roles their degree points at directly are neither plentiful nor well paid.

Who this work suits — and who it doesn't

Life sciences tends to suit people who:

- **Are genuinely curious about how living things work** — the subject has to be interesting on its own terms, because the early pay isn't compensating
- **Are patient and precise** — most experiments fail, and reproducibility is a craft
- **Can tolerate uncertainty** over long periods
- **Are practically dextrous** if aiming at bench work — this is manual skill and it varies
- **Are willing to plan the next step** — the degree is a gateway

It's harder for people who:

- Want **strong early earnings** — the technician tier is where most graduates start
- Need **certainty and closure** — research rarely provides either
- Assume the **degree is the destination**. This is the most common and most costly mistake in this guide.

If some of that second list sounds like you, it isn't a reason to abandon the path — it's a reason to steer toward the parts of it that fit, and toward the regulated end that gives you both protection and pay, which is what the rest of this does.

What else is moving this market

Biotech and pharmaceutical hiring is the growth engine, and it is where the pay is.

Regulatory complexity is expanding, which creates demand for people who understand both the science and the compliance framework — and there are not many.

Lab automation is real but uneven — high-throughput, standardized work automates readily; bespoke and troubleshooting-heavy work does not.

And the academic funding environment is the volatile part, which affects the research career path far more than any technology.

The AI-native advantage — what it actually looks like

The situation: the analysis layer of biology is automating quickly, and the experimental and regulated layers are not. That points to a clear strategy — and being already in it, close to the work, is a real advantage, because you can build this fluency inside a live placement or first job, which is exactly where it counts.

WHAT AN AI-NATIVE LIFE SCIENTIST LOOKS LIKE

1. Own the question, not the pipeline. Running an analysis is no longer a skill. Deciding what analysis would answer the question, and whether the answer is believable, is.

2. Keep the hands. Bench and field capability is the physical protection, and it is what computational-only colleagues lack.

3. Move toward regulated work. Clinical trials, GMP manufacturing, quality and diagnostics all require named accountable humans — and they pay considerably better than academic

research.

4. Verify. Knowing when a generated analysis or a synthesized literature summary is confidently wrong requires understanding the underlying biology well enough to smell it.

5. Understand hypothesis assistance honestly. These tools genuinely help generate candidate explanations. The scientist's value is increasingly in designing the discriminating experiment and interpreting what comes back.

All five are available to you now — not after some future qualification. Turning them into concrete moves is the rest of this guide.

Where a life sciences degree can lead later

Genuinely broad: clinical and professional healthcare routes, industry research and development, manufacturing and process science, quality and regulatory affairs, clinical trial management, diagnostics, environmental and conservation science, forensic science, scientific communication, policy, teaching, and commercial roles in biotech and pharmaceuticals.

The pattern in this index holds with an unusual twist. The further from physical work and from named accountability, the higher the exposure. Pure analysis roles score worst precisely because they have neither. That pattern is the compass for every choice below.

Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: almost none of your exposure is fixed yet. Life sciences is a spread from 4.2 to 7.8, and where you land inside it is decided mostly by things still in front of you. Nearly all of them point the same way: keep your hands, and steer toward regulated work.

If you're still in the degree

Three levers, in order of leverage.

1. The direction is the whole game. The pure-analysis lane — bioinformatics (7.1) and scientific data curation and literature analysis (7.8) — is the exposed end and is furthest into automation already. The protected destinations — field ecology and environmental science (4.2), clinical trials and regulatory affairs (4.6), wet lab and bench research (5.2) — are reached by keeping

physical capability and aiming at named accountability, not by the computational route. If the pull is computational, the highest-leverage move is to pair it with genuine experimental capability rather than treating lab work as the thing to skip — a computational biologist who designs and interprets is protected; one who runs other people's pipelines is not.

2. Regulated exposure, while it's still free. Coursework and placements in regulated environments — GMP, GLP, clinical trial regulation — are rarely taught, highly employable, and reached far more cheaply now than later. This is where the protection and the pay both sit, and almost nobody enters it from a biology degree. Take statistics and experimental design seriously too, as the judgment layer rather than a hurdle.

3. Hands-on experience, early and often. Real lab or field work is the physical protection and the differentiator employers actually assess. That is the experience that separates the protected bench from the exposed analysis layer — and, alongside it, a working theory of what the gateway leads to is worth as much as any single module, because the degree is a gateway, not a destination.

If you've graduated and you're job-hunting

Here the title on the offer matters more than it looks, because the most exposed early-career titles can look like the most advanced ones.

- **"Scientific data curation" and pure "bioinformatics pipeline" roles are the 7.8 and 7.1.** They look tech-forward and well-paid, and they're the automating layer at once. Taking one isn't fatal — but take it as a deliberate stepping-stone with an exit in mind, not as the plan.
- **Aim at clinical trials and regulatory affairs (4.6), environmental and field science (4.2), and wet lab and bench research (5.2).** More protected, and in industry far better paid than the technician tier.
- **Attach to named accountability.** Work where someone signs off a batch, a trial or a diagnostic result inherits real regulatory protection. A seat that's all pipeline output on someone else's data is choosing the exposed end, however advanced it's titled.
- **Chase proximity to the experiment and, in industry, to product.** The three-times pay gap is driven by sector, not ability — a regulated or industry role beats a higher-status computational seat funded by grants and being automated underneath you.

The AI-native move, from where you stand

You can't undo the tasks that are already automating, but you can be the scientist who directs these tools rather than competes with them. The five capabilities above — own the question, keep the hands, move toward regulated work, verify, understand hypothesis assistance

honestly — are all available right now, inside a current placement or a first job. The most powerful for someone already in it is verification: be the person who catches the confidently-wrong analysis or the synthesized literature summary that's subtly off. That only works on biology you actually understand — which is exactly the judgment the protected end is built on, and you can start building it this term, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in life sciences is built here, in these habits and choices, rather than handed to you. The first list is for anyone still in the degree; the second is for the job hunt itself.

During the degree — build toward the protected, regulated end

- **Aim past the technician tier from the start.** The protection *and* the pay both live at the regulated end (field ecology & environmental science 4.2, clinical trials & regulatory affairs 4.6, wet lab & bench research 5.2), not in the technician tier or the pure-analysis lane (bioinformatics 7.1, scientific data curation 7.8). Know which end each choice points at.
- **Keep the hands — get real lab or field experience early and often.** This is the physical protection and the differentiator employers actually assess. If the pull is computational, pair it with genuine experimental capability rather than treating lab work as the thing to skip.
- **Move deliberately toward regulated work.** Coursework and placements in clinical trials, GMP manufacturing, quality and diagnostics are under-applied to, protected by named accountability, and pay far above the technician tier — and almost nobody enters them from a biology degree.
- **Take statistics and experimental design as judgment, not a hurdle.** Deciding what analysis would answer the question — and whether the answer is believable — is the layer that survives; running the pipeline is not.
- **Get fluent with AI-assisted analysis as a scientist, not a shortcut.** Learn to catch the confidently-wrong analysis or literature summary — verification only works on biology you actually understand.
- **Feed curiosity and adaptability on purpose.** The pipelines and platforms will turn over repeatedly; the scientist who re-tools fastest, and stays genuinely curious about how living systems work, compounds.

- **Keep the frame wide, and have a working theory of the gateway.** The degree is a gateway, not a destination — vet professional programs, specialist master's, industry and regulated routes, and know what the next step is before you finish, not after.

During the job hunt — aim for the protected lanes, don't just apply

- **Target the protected lanes on purpose.** Clinical trials and regulatory affairs (4.6), environmental and field science (4.2), wet lab and bench research (5.2) — not "any bioinformatics or data-analysis role," which points straight at the 7.1 and 7.8.
- **Read the title, not just the offer.** Scientific data curation and literature analysis (7.8) and pure pipeline bioinformatics (7.1) are the automating layer. If you take one, take it as a deliberate stepping-stone with an exit in mind, not as the plan.
- **Don't grab a job just because it's on offer in an exposed lane.** A role being available doesn't make it safe. A lab still hiring heavily for pure pipeline work or literature curation may simply be behind the curve — and that's exactly the rung that gets cut first once it catches up. Availability can be a lagging indicator, not a green light.
- **Prefer proximity to the experiment and to named accountability.** A seat at the bench, in the field, or where someone signs off a batch, a trial or a diagnostic result puts you close to the protected end — and, in industry, close to the pay the technician tier lacks.
- **Ask the employer where AI is heading in the role.** Whether they're rebuilding the job around directing and verifying these tools, or waiting to be forced to, tells you if the seat grows or is quietly being hollowed out.
- **Weigh the sector and the trajectory over the starting number.** The three-times gap between the technician tier (around \$47,000) and industry (mid \$140,000s to low \$160,000s) is driven by sector, not ability — a protected, regulated lane beats a higher-status computational seat being automated underneath you.

Go deeper — further reading

- **Biological Technicians — Occupational Outlook Handbook** — US Bureau of Labor Statistics. The pay and growth reality for the role most biology bachelor's graduates enter first. Read the **Medical Scientists** and **Environmental Scientists** pages alongside it — the contrast is the useful part.
- **BLS Life, Physical, and Social Science Occupations** — the full set, with growth projections and median pay for every track in this guide. Free and comparable.

- **US Life Sciences Salary Guide 2026** — industry compensation by function: manufacturing, quality, regulatory, commercial. Note it's published by a staffing firm, so read it for the shape of the market rather than as neutral — but the gap it shows against academic pay is the important finding.

The counterargument — read this one:

The strongest case against our 7.1 for bioinformatics is that we are scoring the *technique* rather than the *science*.

On this view, a computational biologist is not a data analyst who happens to work on biology — they are a scientist whose instrument is a computer. The judgment about which analysis is meaningful, which result is artefactual and which finding is worth pursuing is scientific judgment, and it is no more automatable than a bench scientist's. Automating the pipeline no more replaces the computational biologist than automating a sequencer replaced the molecular biologist.

Where we land: we think this is right about the best computational biologists and wrong about the median role. Our score reflects the work as most commonly performed — running established pipelines on data someone else generated, to answer questions someone else framed. The scientists who design and interpret are protected. The distinction is judgment and ownership, and it happens to be exactly the distinction that separates tracks in every other guide in this index.

What we're watching: whether computational roles increasingly carry experimental design responsibility. If they do, the score should come down.

Bottom line

Life sciences contains an inversion worth understanding: the computational end is more exposed than the bench. That's the whole reason being already in it is not a problem — the decisive moves were always going to happen from here.

Bioinformatics scores 7.1 against wet lab research at 5.2 — and the gap is widening rather than closing. The reason is structural: computational biology takes structured inputs and produces analysis, which is the exposed profile in every field we score. The biology is the subject; the shape of the work is computation.

But the more important finding is financial. A life sciences bachelor's most commonly leads to technician work at around \$47,000, while the same field in industry averages mid \$140,000s to

low \$160,000s — a gap of roughly three times, driven by sector rather than by ability.

So the useful advice isn't a verdict on the degree — it's a direction. Keep hands-on capability rather than trading it for computational skill. Steer toward regulated work — clinical trials, quality, manufacturing, diagnostics — because it is both the protected version and the well-paid one, and almost nobody applies to it from a biology degree. Take statistics seriously as judgment rather than as a hurdle. Have a working theory of what the gateway leads to. And be honest about the academic path, where far more people are trained than there are permanent positions.

And one thing worth saying if you simply find living things fascinating. **That's a good reason to be here, and it's not sufficient on its own. The people who do well from a life sciences degree are almost always the ones who worked out early what it was a gateway to.** The good news is that working it out is a move you can still make from exactly where you are.

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which end are you pointed at — the hands-on, regulated end (field ecology 4.2, clinical trials & regulatory affairs 4.6, wet lab 5.2) or the pure-analysis lane (bioinformatics 7.1, data curation 7.8)? And is it the one you'd choose on purpose? 2. If the pull is computational: are you pairing it with genuine experimental capability, or treating lab work as the thing to skip? 3. Do you have real hands-on lab or field experience yet — or only analysis? If it's the latter, what's the nearest way to change that?

The move from here

4. What's the single most protected seat you could realistically aim for from where you are now — regulated work, the bench, field science — and what would make you a credible candidate for it? 5. Do you have a working theory of what the degree is a gateway to — medicine, a specialist master's, industry, regulated work? This is the single most important question in this guide. 6. If you're job-hunting: are you aiming at a protected, regulated lane on purpose, or applying to whatever's open? Name the three roles you're actually chasing.

The AI question, practically

7. Of the five capabilities — own the question, keep the hands, move toward regulated work, verify, understand hypothesis assistance honestly — which one could you visibly demonstrate within the next few months? 8. Can you point to a moment where you caught a generated analysis or literature summary that was subtly, confidently wrong? If not, where's the easiest place to start building that skill?

Testing it against reality

9. **The most valuable single action:** get real lab or field experience, and find someone three to five years past a biology degree — ask what they actually do and what they earn. 10. If the first offer that comes is a pure data-curation or pipeline role, what's your plan — take it as a deliberate stepping-stone with an exit in mind, or hold out for something closer to the bench or a regulated environment? Decide before the offer, not after.

This is analysis and informed judgment about a fast-moving situation. For life sciences specifically, we've flagged that our score reflects the median computational role rather than the best ones, and that the field's most significant practical issue — the gap between academic and industry compensation — sits entirely outside what our six factors measure.

Scores for 2023 and 2025 are retrospective reconstructions using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Life Sciences

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
--------	--------

0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
--------	--------

0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
--------	--------

0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

FIELD ECOLOGY & ENVIRONMENTAL SCIENCE — 4.2 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.0	5.7	6.4	2.24
Entry-path erosion	15%	3.0	3.4	3.8	0.57
Physical / embodied	15%	8.5	8.5	8.5	0.22
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	5.5	5.5	5.5	0.45
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	4.21 → 4.2

CLINICAL TRIALS & REGULATORY AFFAIRS — 4.6 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.2	6.2	6.9	2.42
Entry-path erosion	15%	3.2	3.6	4.0	0.6
Physical / embodied	15%	3.5	3.5	3.5	0.97
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.61 → 4.6

WET LAB & BENCH RESEARCH — 5.2 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	6.6	7.6	2.66
Entry-path erosion	15%	3.5	4.0	4.4	0.66
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	2.5	2.5	2.5	0.75
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	5.2 → 5.2

LAB TECHNICIAN & DIAGNOSTICS — 5.8 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.7	7.9	8.9	3.11
Entry-path erosion	15%	4.0	4.5	5.0	0.75
Physical / embodied	15%	7.5	7.5	7.5	0.38
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	5.5	5.5	5.5	0.45
Judgment under novelty	10%	5.5	5.5	5.5	0.45
				Total	5.82 → 5.8

BIOINFORMATICS & COMPUTATIONAL BIOLOGY — 7.1 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.9	6.8	8.6	3.01
Entry-path erosion	15%	4.5	5.2	5.8	0.87
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	7.1 → 7.1

SCIENTIFIC DATA CURATION & LITERATURE ANALYSIS — 7.8 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	7.7	9.2	3.22
Entry-path erosion	15%	5.5	6.2	6.8	1.02
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	4.5	4.5	4.5	0.82
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	7.82 → 7.8

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Field ecology & environmental science

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	5.0	3.0	1.4	3.6
2025	5.7	3.4	1.4	3.9
Fall 2026	6.4	3.8	1.4	4.2

Movement decomposition: automatability contributed **+0.49** and entry-path erosion **+0.12**, for a total of **+0.6** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.

- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.